

# Sonny Hernandez

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## EDUCATION

**TEXAS A&M UNIVERSITY:** College Station, TX Bachelor of Science in Computer Engineering, Expected graduation MAY 2027

## WORK EXPERIENCE

### Professor Assistant Leader (Python)

AUG 2023

- Assisted the professor in classes of Engineering Computation specifically using the computing language Python
- Conducted interactive Python workshops to enhance students' understanding and proficiency in the language.
- Evaluated student assignments and provided constructive feedback to facilitate their skill development.
- Maintained a positive and engaging classroom environment that encouraged active participation and learning.

### Professor Assistant (Physics Engineering Lab: Mechanics)

FEB 2024

- Assisted the professor in classes of physical mechanics.
- Conducted interactive kinematic workshops to enhance students' understanding and proficiency in kinematics and physical analysis.
- Evaluated student assignments and provided constructive feedback to facilitate their skill development.

### Machine Learning Researcher - Texas A&M University FIGURES Program

MAY 2025

- Contributed to a **PhD** research topic, "*Detection of Nuts and Bolts in Complex Environments*," applying computer vision techniques for classification and detection.
- Created and preprocessed a custom dataset; implemented object detection models including **YOLOv8** and a custom **TensorFlow** pipeline.
- Designed a vision-based detection system using Python, OpenCV, and deep learning frameworks, with deployment planned for **NVIDIA Jetson Orin**.
- Benchmarked YOLOv8 against a custom TensorFlow pipeline, evaluating trade-offs in accuracy, speed, and efficiency.

## PROJECTS

### Mars Rover Simulation Lesson (TA lesson to students)

AUG 2023

- In this multi-week project, students engaged in hands-on learning to simulate the design and operation of a Mars rover. The project was divided into two phases: rover design and rover programming.
- Students worked in teams to design and build a LEGO Mars rover. Students had to consider factors like mobility, stability, and instrument placement.
- After completing their rover construction, students learned how to program the LEGO EV3 robot using **Micro Python**. They programmed their rovers to perform tasks such as navigating obstacles, collecting data, and simulating Martian terrain exploration.

### ESP-32 IoT Pager Device

MAY 2025

- Built a custom pager with ESP32, OLED display, and Li-Po battery.
- Developed Flask server on Raspberry Pi for remote notifications via **ngrok** tunnel.
- Implemented power management features using deep sleep and TP4056 charging module for extended runtime.
- Integrated buzzer and Python script to alert user when computer vision model training completed.

## CERTIFICATION & TRAINING

### Future of Work Institute, Deloitte

- 12-hour certification gained insight into the expected evolution of the workplace over the next decade and how to navigate difficult workplace conversations.

**Technical Skills:** Python, C++, TensorFlow, YOLOv8, Docker, Jetson Orin, Raspberry Pi, AutoCAD, Verilog, Zybo-Z7 FPGA Board